Class 3: Thermal Expansion

In this class we will look at a particular material property – thermal expansion. One of the reasons we will look at this property in this class, early in this course, is that it lends itself to considerable analysis and understanding in a rather concise manner. Therefore, by looking at how a material is modeled to explain thermal expansion, we get a preview of the type of approach we will follow in the rest of this course. Unlike thermal expansion, some of the other properties we will look at later in this course, will require considerably more involved analysis and more extensive development of the related background material. While we will be able to address thermal expansion in a single class, some of the other properties will require us to examine multiple models and develop the related background material over several classes.

Several day to day products use thermal expansion. A bimetallic strip which consists of two metals with different coefficients of thermal expansion, will bend or straighten, based on the temperature. Bimetallic strips have been used in iron boxes, electric kettles, and refrigerators. A schematic showing the functioning of the bimetallic strip is shown in Figure 3.1

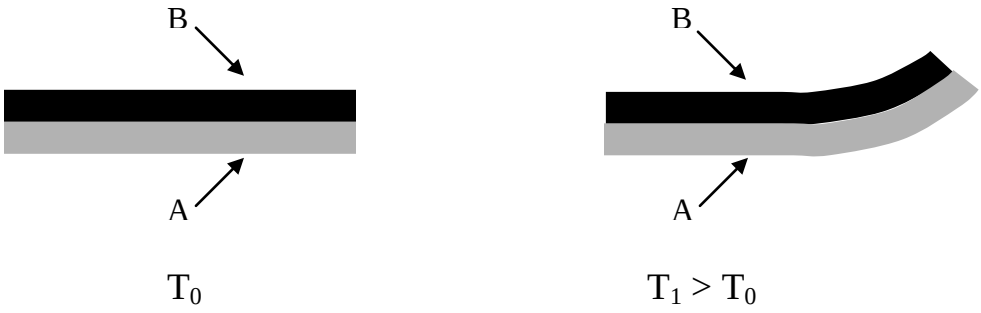


Figure 3.1: Schematic of a bimetallic strip, made of metals 'A' and 'B'. At a lower temperature T_0, the two metals have the same length. At a higher temperature T_1, metal 'A' has expanded more than metal 'B', and this results in the strip bending to accommodate the different lengths.

While thermal expansion is made use of to our advantage in many places, there are several situations where thermal expansion is not a welcome phenomenon. In any application where the dimensional stability of components is important, thermal expansion has to be avoided. An example where this is evident is shown in Figure 3.2, where two pipes are connected and the joint between them is sealed using a gasket. When the temperature of this setup is lowered, the two pipes as well as the gasket can shrink and this can result in the seal between the pipes getting broken.

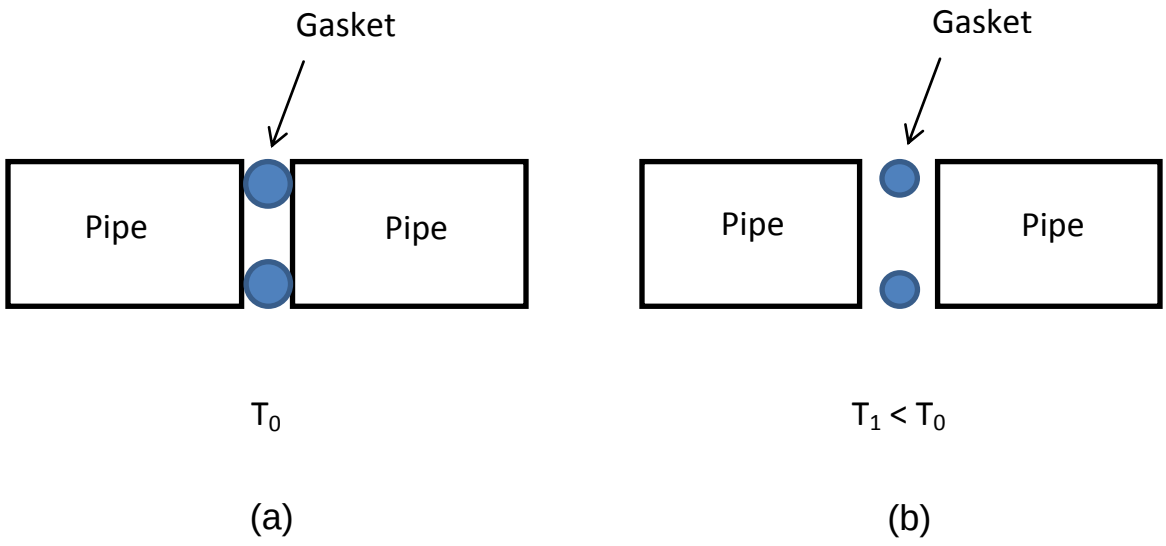
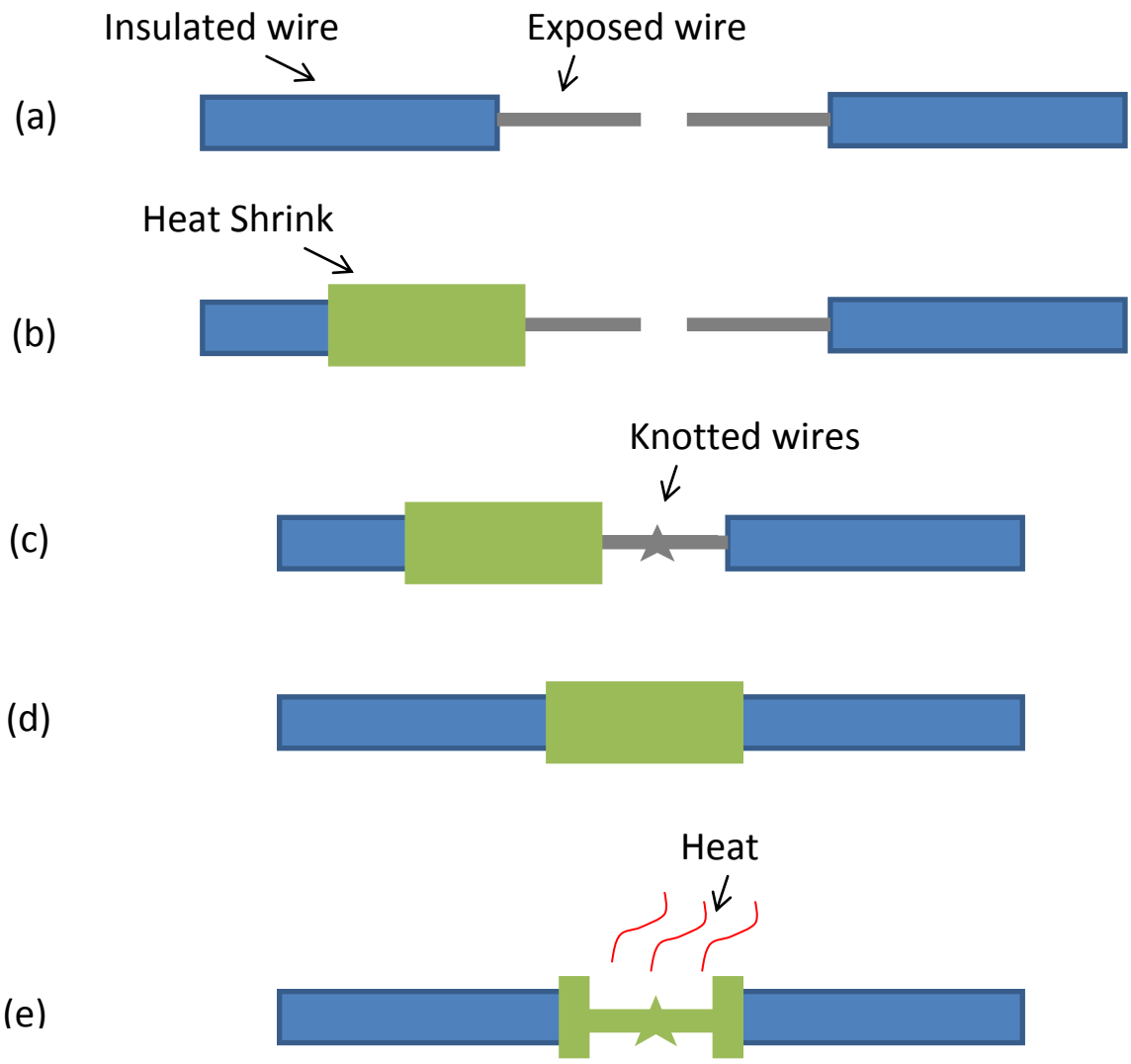


Figure 3.2: (a) Schematic showing the side view of two pipes connected using a gasket for sealing purposes, at temperature T_0. (b) As the temperature of the joint is lowered to T_1, the seal is broken because the pipes as well as the gasket have shrunk.

Incidentally, while most materials expand upon heating, there are some materials that are designed to shrink upon heating. These materials are commercially referred to as 'Heat Shrinks'. They are polymer based materials that operate in one of two ways. In some cases, the crosslinking process is deliberately stopped in early stages and the material is supplied in that state. When such a material is heated, the crosslinking process proceeds to completion. During crosslinking the molecules are pulled closer to each other, and as a result the overall material shrinks. Another method to accomplish the same is to supply the material with additives that can outgas from the material on heating. When the additives leave the material, the remaining material shrinks.

One of the commercial uses for this type of a material is to enable electrical insulation of joints of exposed wires. A schematic showing the application of a heat shrink material to enable electrical insulation, is shown in Figure 3.3. Heat shrink is available in the form of hollow tubes and during its application one of the exposed wires is slid through an appropriate length of the tube. The length of the tube used should be adequate to cover the length of the exposed wires as well as to enable a limited overlap with the insulated parts of the wires. The exposed wires are then knotted to each other. The heat shrink tube is then slid on top of the knotted joint and is heated, typically using hot air. The tube shrinks and forms a good insulating cover over the knotted joint.



Animation of the above is shown in the next page

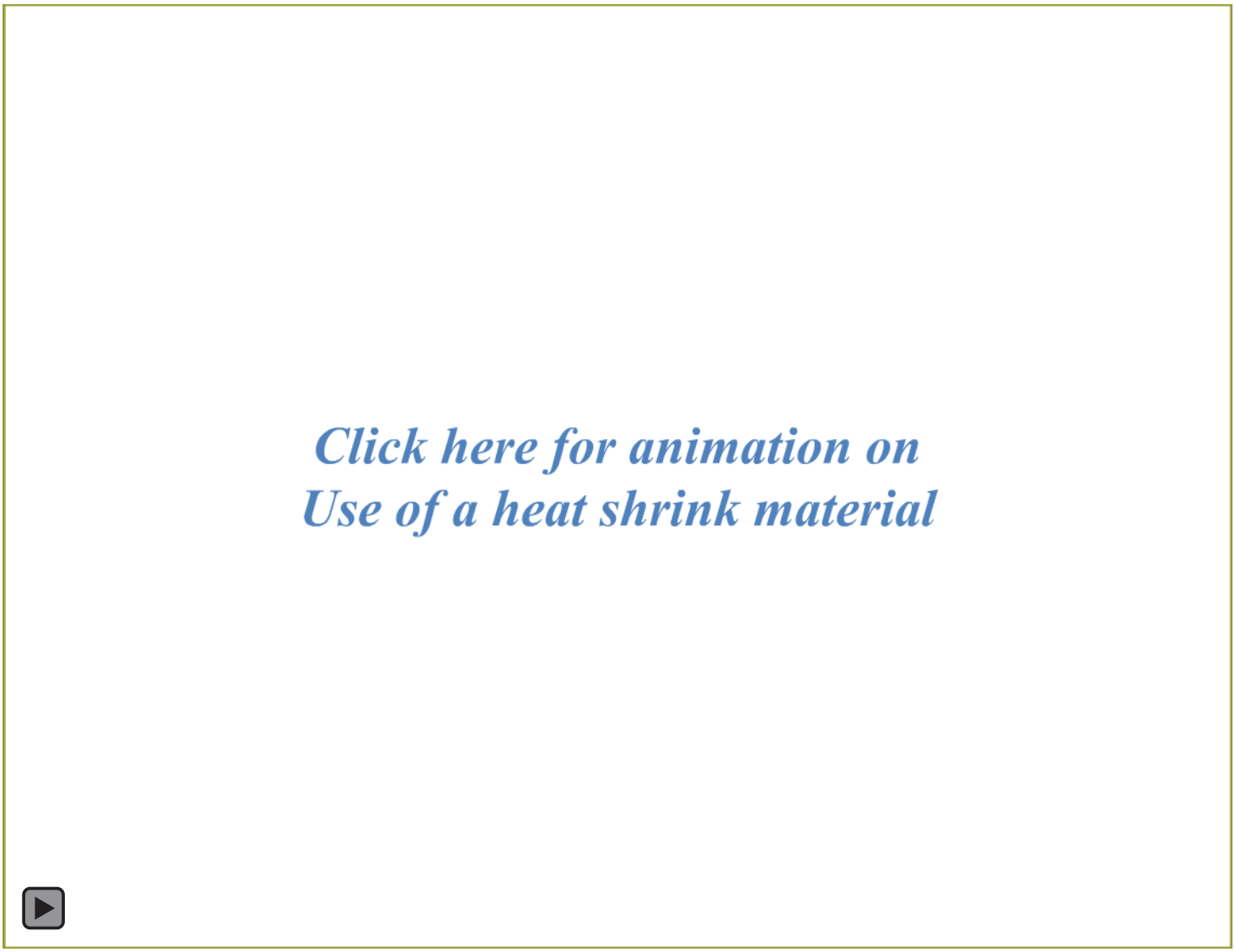
Figure 3.3: Use of a heat shrink material (a) Two exposed wires that need to be joined and insulated. (b) Heat shrink, in the form of a hollow tube, is slid over one of the wires. (c) The wires are knotted together. (d) The heat shrink is slid on top of the knotted wires. (e) Heat is applied and the material shrinks to form an insulating cover over the exposed wires and the knot. Since the material shrinks till it is physically stopped, it takes the shape of the knot in the location of the knot.

Although we have briefly looked at heat shrink materials in the discussion above, as indicated earlier, most materials expand on heating. In general, if the material has a length L_0, at temperature T_0, and a length L_1 at temperature T_1, the lengths can be related using the expression:

$$L_1 = L_0 + L_0\alpha(T_1 - T_0)$$

Where 'α' is the linear coefficient of thermal expansion

Animation of figure 3.3: Use of a heat shrink material



The question is, why does thermal expansion occur?

Common perception is that materials expand on heating because atoms in a solid vibrate more, or have larger amplitudes of vibration, at higher temperatures. While it is true that atoms in a solid do vibrate with larger amplitudes at higher temperatures, this by itself cannot result in thermal expansion. The reason is that if the mean position of the vibration of the atoms does not change, then it does not really matter if the amplitude of the vibration has gone up – on average the inter-atomic distance will be the same and the material will not expand. We therefore need to look deeper to understand why thermal expansion occurs.

To understand thermal expansion, let us take a simplified example of a one dimensional, ionically bonded, material, and see how it comes together starting from individual atoms. By examining this process and its implications, we will be able to get a clearer understanding on why thermal expansion occurs.

Note: NaCl has a 'rock salt' crystal structure which has the larger Cl^- ions occupying FCC sites, and the smaller Na^+ ions occupying the octahedral sites in the FCC structure. In the analysis below we simplify the structure and look at it from a 1 Dimensional perspective.

Let us consider a one dimensional crystal of NaCl.
Since it is an ionic crystal, it consists of Na^+ and Cl^- ions. To begin with we will look at Na and Cl separately and then put them together to form the crystal.
When an independent atom of Na, becomes a univalent ion, the reaction can be written as follows:

$Na \rightarrow Na^+ + e^-$ +5.1 eV

Ionization energy of +5.1 eV has to be provided per atom to ion transition, and this raises the energy of the system.

Independently, let us consider the formation of a Cl^- ion:

$Cl + e^- \rightarrow Cl^-$ -3.7 eV

Electron affinity of 3.7 eV per atom to ion transition, is released in this process.
The net energy required to create an Na^+ ion and a Cl^- ion, can be obtained from the sum of the above two reactions, and is +1.4eV per pair of ions produced as indicated below:

$$E = 1.4 \times 1.6 \times 10^{-19} \text{ J}$$

This energy does not depend on the inter-ionic distance between the two ions, and is therefore as shown in the dotted line in Figure 3.4(a). Since this is always a positive number, this alone will not enable the stable formation of the NaCl ionic crystal.
To consider the energy that results due to the electrostatic interaction between the ions, let us consider the possibility that they are infinitely separated from each other initially,

and are therefore not interacting with each other, and then steadily brought closer. The energy due to the coulombic attraction between the ions is given by the equation below:

$$E = \frac{-1}{4\pi\varepsilon_0}\frac{e^2}{R}$$

Where 'R' is the inter-ionic distance, 'e' is the charge of an electron, and ε_0 is the permittivity of free space. This interaction is at an atomic, or ionic level. The energy associated with this interaction, is plotted as a function of inter-ionic distance in Figure 3.4(b). This energy is negative, and if only this were to exist, the ions would collapse into each other.

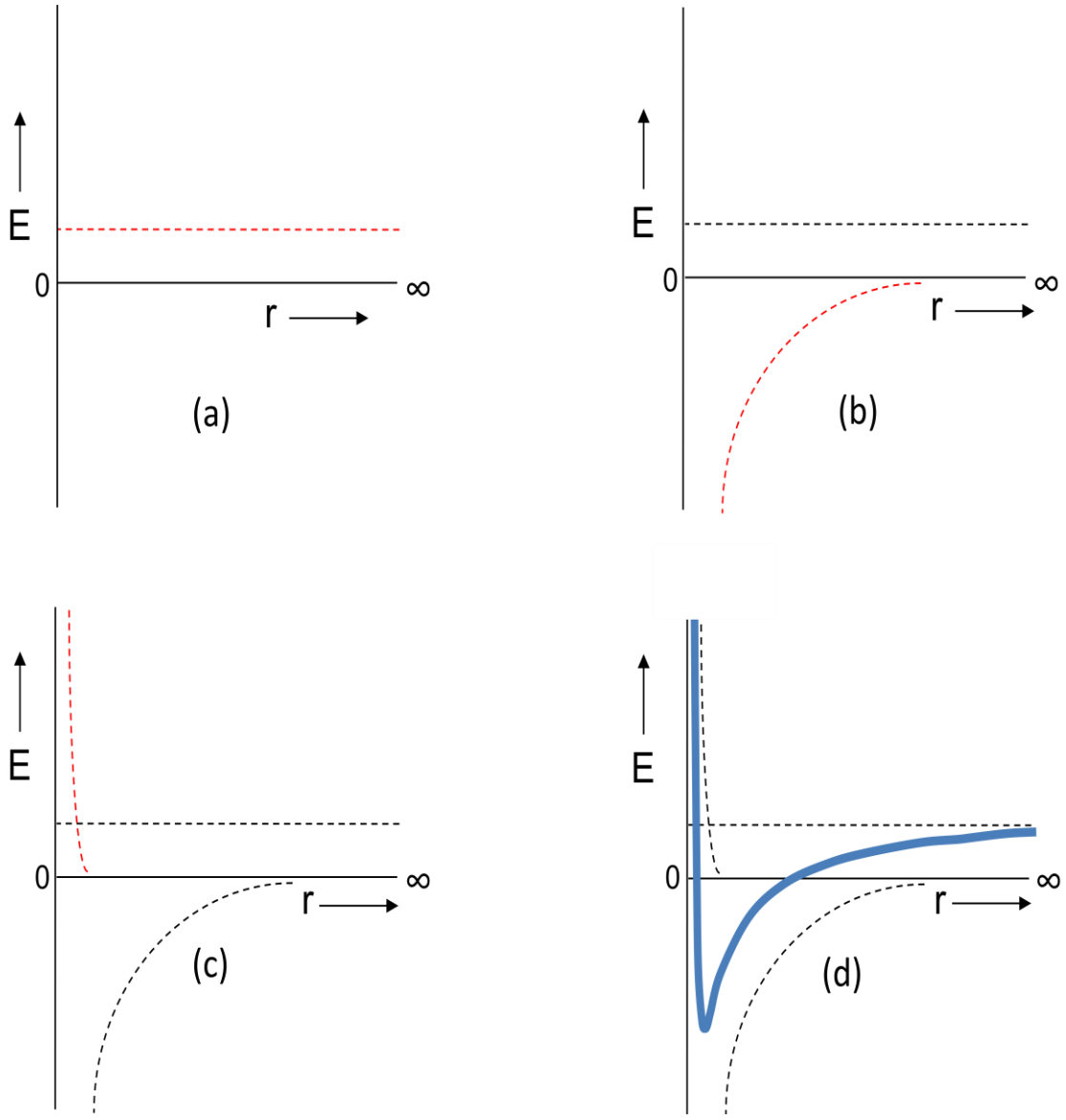
As the ions get significantly close to each other, the electron clouds, especially the outer electron shells of the ions, begin to overlap. This forces electrons to move to higher energy levels in keeping with the Pauli's Exclusion principle. This interaction is of a sub atomic level and causes the energy of the system to increase. The dependence of the energy on inter-ionic distance, due to this interaction, is given by the equation below, and is shown in Figure 3.4(c).

$$E = \frac{B}{R^n}$$

This energy is positive and raises sharply as R decreases since its dependence on R is as R^{10}, i.e. n is usually of the order of 10.

The equation below gives the sum of the energies involved.

$$E = \frac{-1}{4\pi\varepsilon_0}\frac{e^2}{R} + \frac{B}{R^{10}} + 1.4 \times 1.6 \times 10^{-19}$$



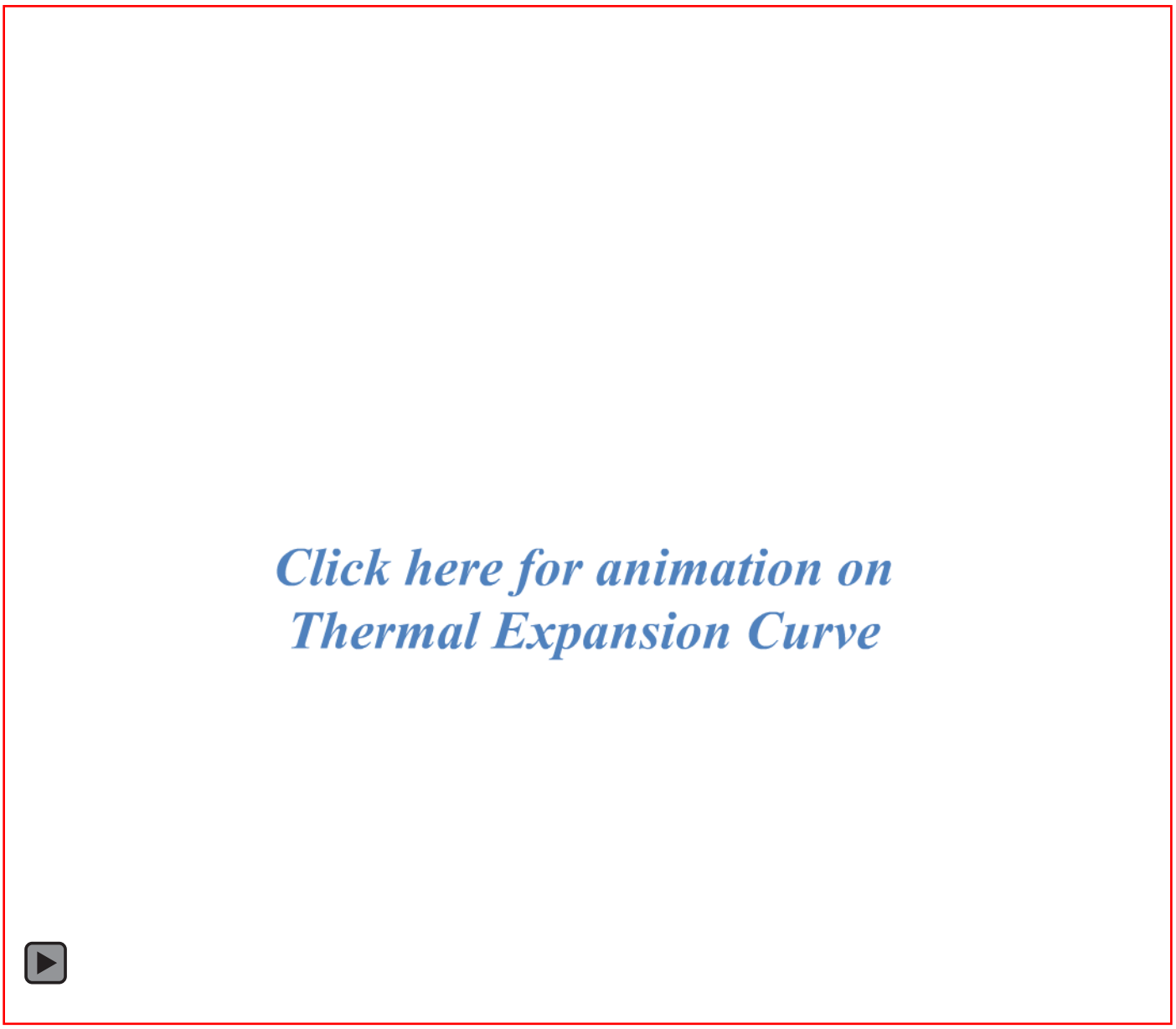
Animation of the above is shown in the next page

Figure 3.4: Schematic showing the contributions to the energy involved in the formation of a one dimensional NaCl crystal. (a) Dotted red line shows the sum of Ionization energy and Electron Affinity. (b) Dotted red curve results from the coulombic attraction between the ions. (c) Dotted red curve shows the repulsion due to the overlap of electron clouds as the ions get closer. (d) The solid blue line shows the resultant energy as a function of inter-ionic distance

Figure 3.4(d) shows the sum of all of the energies as a function of R, in the form of the solid curve.

The sum of the energies involved passes through a minimum E_0, at which point the inter-ionic separation is a single fixed value of r_0, as shown in Figure 3.5.

Animation of figure 3.4: Thermal Expansion Curve



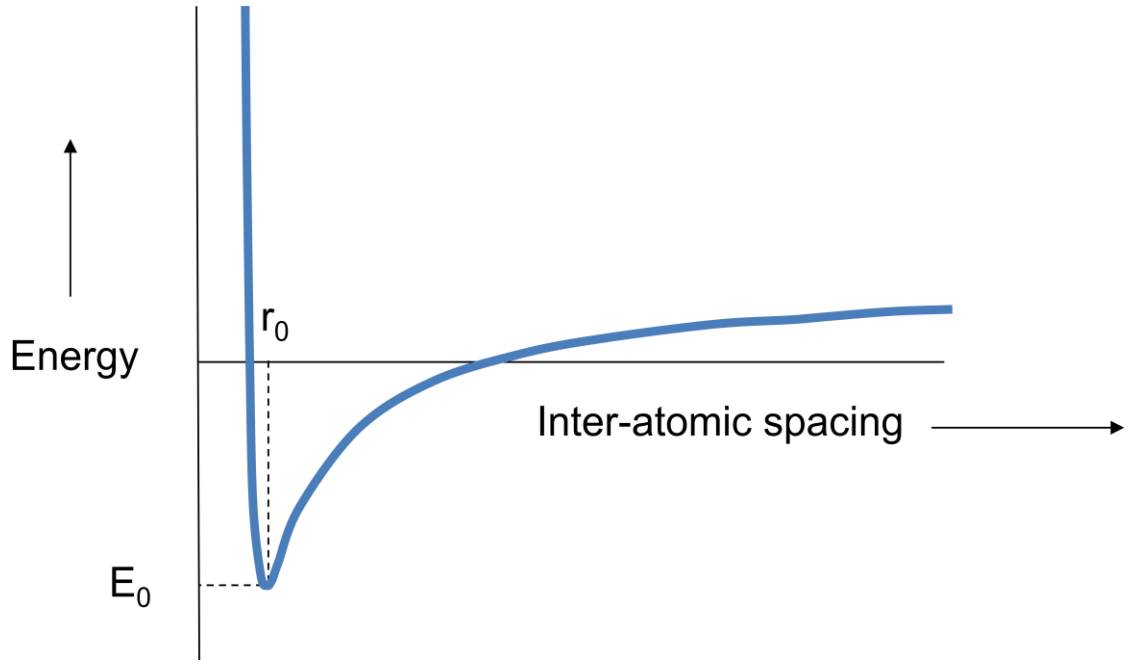


Figure 3.5: The minimum energy E_0, and equilibrium inter-ionic spacing r_0, at 0 Kelvin.

If the inter-ionic separation is fixed, it means the atoms are stationary, and this can occur only at 0 Kelvin. Therefore the atoms are at their minimum energy E_0, and inter-ionic separation r_0, at 0 Kelvin.

As the temperature of the system is raised to T_1, and then further to T_2, the solid becomes consistent with the higher temperatures by having the ions gain energy, and vibrate with increased amplitudes. Therefore, as shown in Figure 3.6, at T_1, when the energy is E_1, the ions are able to vibrate between the positions 'A' and 'B'. Similarly, at T_2, when the energy is E_2, the ions are able to vibrate between the positions 'C' and 'D'.

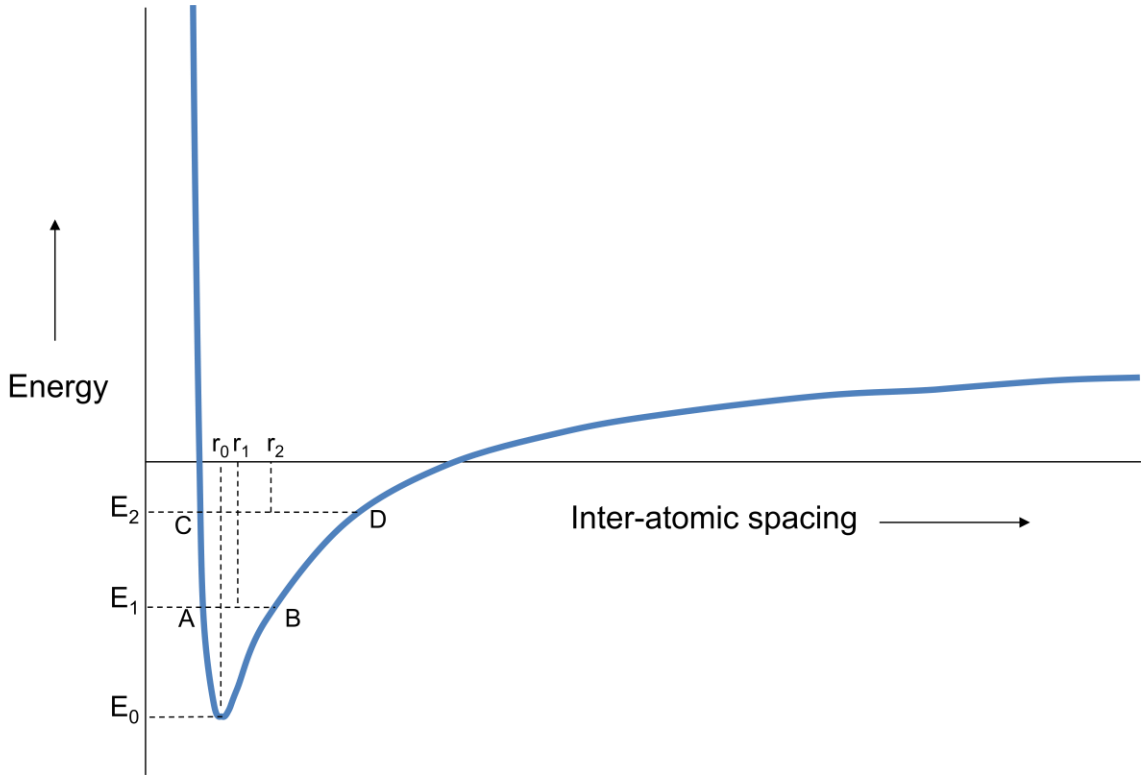


Figure 3.6: Effect of increasing temperature on the vibration of the ions and hence on the inter-ionic separation. The system is raised to temperature T_1, and then further to T_2. At T_1, when the energy is E_1, the ions are able to vibrate between the positions 'A' and 'B'. Similarly, at T_2, when the energy is E_2, the ions are able to vibrate between the positions 'C' and 'D'

It is important to note from the Figure 3.6 that, the midpoint between 'A' and 'B', which represents the mean inter-ionic distance at T_1, is r_1 which is greater than r_0. This is a direct result of the fact that the solid curve in Figure 3.6, is asymmetric. The solid curve is asymmetric as a direct result of the form of the three terms (or curves) that add up to result in the solid curve. The manner in which the three terms in Equation depend on R, and therefore the manner in which they display their influence in the sum, results in the solid curve being asymmetric.

It is therefore important to note that thermal expansion occurs as a direct result of the fact that the 'E Vs R' curve is asymmetric.

A hypothetical solid, in which the E Vs R curve is exactly symmetric, is shown in Figure 3.7. In this case when the temperature is raised, E increases, and the amplitude of vibration of the ions increase, but there is no thermal expansion since the mean inter-ionic distance is exactly the same at all temperatures.

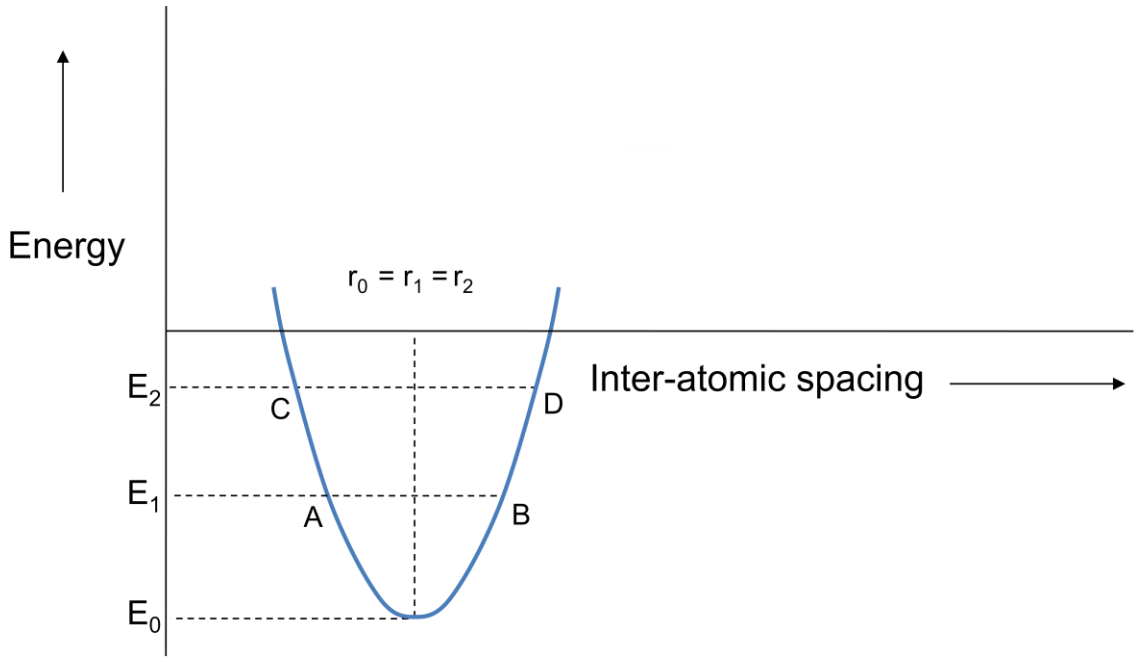


Figure 3.7: A hypothetical solid, in which the E Vs R curve is exactly symmetric. In this case when the temperature is raised, E increases, and the amplitude of vibration of the ions increase, but there is no thermal expansion since the mean inter-ionic distance is exactly the same at all temperatures

It is also important to note that while the general approach used here can be extended to many systems, the exact shape of the resultant curve obtained will depend on the details of the specific system. In some cases, such as ceramic materials, the resultant curve will have a very deep and narrow trough, such materials will have a very low coefficient of thermal expansion. In other cases, the resultant curve will have a shallow and wide trough, which will cause the material to display a high coefficient of thermal expansion.

In summary, in this class a model has been built to describe an ionic solid, and a property of the solid, thermal expansion, has been explained on the basis of this model. This model uses interactions at the atomic and subatomic levels to explain macroscopic phenomena. A similar approach, with varying levels of details will be used in later classes to explain other material phenomena.